

SERKAN BERK BILGIC

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SUMMARY

Software Engineer with backend-leaning full-stack experience building secure, scalable products across the SDLC. Proficient in Python and Java, designing HTTP/REST APIs, multi-tenant data access, and production-grade reliability patterns (retries, idempotency, structured logging). Comfortable in Agile teams with CI/CD and security constraints.

EDUCATION

Koc University

B.Sc. Computer Engineering + Economics (Double Major), Full Scholarship; GPA: 3.8/4.0

Istanbul, Turkey

Jun 2024

University of Mannheim

Erasmus Study (Economics); GPA: 3.62/4.0

Mannheim, Germany

Feb 2024-Jun 2024

WORK EXPERIENCE

Datapad

Software Engineer (Full Stack / AI)

Istanbul, Turkey (Hybrid)

Feb 2025-Present

- Built and shipped end-to-end product features for a multi-tenant analytics platform using Next.js (TypeScript) and Python services, collaborating closely with product and design to deliver polished user experiences from concept to production.
- Designed and implemented secure REST APIs with OAuth2/JWT authentication, role-based access control (RBAC), and strict tenant data isolation, ensuring consistent authorization boundaries and safe multi-tenant access patterns in prod.
- Co-built Echo (get-echo.ai) by implementing AI agent workflows with an Agent SDK, building agent tools, TypeScript orchestration logic, and prompt-layer guardrails to reliably execute customer workflows end-to-end.
- Developed tenant knowledge retrieval with vector embeddings and retrieval pipelines, improving response quality by tightening context assembly, reducing irrelevant recall, and refining tool routing for more accurate answers.
- Implemented tenant data ingestion with Inngest (async jobs, retries, idempotency), storing data in Neon (Postgres) with Redis caching, and improving reliability on Vercel via structured logging, defensive error handling, and monitoring-ready instrumentation at scale. Worked with enterprise customers including Hipicon, IKEA, LC Waikiki, Les Benjamins.
- Migrated Datapad AI Data Analyst backend infrastructure from GCP to Railway to simplify deployments and improve operational velocity, and built text-to-SQL and text-to-Mongo agents to enable natural-language querying.

Siemens

Working Student, Software Engineering

Istanbul, Turkey

Mar 2023-Aug 2023

- Developed and maintained Java services for a network planning tool in a large corporate environment; delivered bug fixes and features in an Agile team. Contributed to design discussions, code reviews, and testing practices; supported operational stability and troubleshooting of production issues. Used Java, Spring Boot, Docker, Azure DevOps.

Amadeus

Software Development Engineer Intern

Istanbul, Turkey

Jun 2022-Aug 2022

- Built an internal admin panel using Java, Spring Boot to help support agents track issues & resolve customer tickets faster.

Tor.app (Transkriptor & Spektor)

Growth, Data & Product Specialist

Istanbul, Turkey (Remote)

Mar 2024-Jan 2025

- Benchmarked STT/TTS vendor APIs by locale and recommended model switches based on accuracy, latency, and cost across web and mobile. Ran pricing and conversion experiments; improved conversion up to 40% in selected locales.

SELECTED PROJECTS

Stock AI

React Native iOS App

- Built and shipped an iOS app that analyzes stock charts and generates AI-assisted explanations and scenarios.

Backsmith.ai

Web Prototype

- Built an AI LinkedIn writer prototype and contributed to content automation workflows (demo available on request).

ADDITIONAL

Hard Skills: Python, Java, TypeScript/JavaScript, SQL, Spring Boot, FastAPI, Node.js/Next.js; HTTP, REST

APIs, OAuth2/JWT, webhooks, RBAC, caching, PostgreSQL, MongoDB, Redis, Docker, Git, CI/CD, Vercel, GCP, Azure DevOps

Soft Skills: Teamwork, ownership, MVP judgment, clear communication, cross-functional collaboration, problem-solving, bias for action, attention to detail, reliability.